Software Engineering — Lecture 04: Requirements Engineering

بنك الأسئلة — الفصل الرابع: Requirements Engineering

Part 1 — MCQ

1. What is requirements engineering (RE)? high

ات (RE)؟

A. The process of writing and testing the code that implements the system services
B. The process of finding out, analyzing, documenting and checking the services and constraints of a system
C. The process of estimating the cost, schedule and staffing of a software project
D. The process of deploying the system and training its users

ة الصحيحة: B. The process of finding out, analyzing, documenting and checking the services and constraints of a system

يه صح: اكتشاف وتحليل وتوثيق والتحقق من الخدمات والقيود. الاختيار الأقرب (كتابة الكود) غلط لأنه تنفيذ مش متطلبات.

جع: Chapter 4 — Introduction

2. Which statement best describes USER requirements? high

دم (User requirements) بشكل أعلى؟

A. Statements, in natural language plus diagrams, of the services the system is expected to provide to its users and the constraints under which it must operate
B. Detailed descriptions of the system's functions, services and operational constraints that define exactly what is to be implemented
C. Formal mathematical models of the system behavior written for the developers
D. Descriptions of the system architecture and software design decisions

ة الصحيحة: A. Statements, in natural language plus diagrams, of the services the system is expected to provide to its users and the constraints under which it must operate

يه صح: متطلبات المستخدم بلغة طبيعية ورسومات وعلى مستوى عالٍ. الاختيار الأول وصف لمتطلبات النظام (System requirements) وهو الأ

جع: Chapter 4 — Introduction

3. Which statement best describes SYSTEM requirements? high

ظام (System requirements)؟

A. High-level abstract statements of services written in natural language for the client managers
B. More detailed descriptions of the system's functions, services and operational constraints that should define exactly what is to be implemented
C. Statements of the needs of the end-users written without any technical detail
D. A short early study of whether the system is technically and financially feasible

ة الصحيحة: B. More detailed descriptions of the system's functions, services and operational constraints that should define exactly what is to be implemented

يه صح: متطلبات النظام أكثر تفصيلًا وتحدد بالظبط إيه اللي هيتنفذ. الاختيار الأول يخص User requirements، والأخير يخص Feasibility study

جع: Chapter 4 — Introduction

4. The system requirements document is sometimes called a: high

يقة متطلبات النظام بتتسمى أحيانًا:

A. Feasibility study
B. User requirements definition
C. Functional specification
D. Software design document

ة الصحيحة: C. Functional specification

يه صح: الكتاب بيقول إن System requirements document ممكن تتسمى Functional specification. User requirements definition

جع: Chapter 4 — Introduction

5. Why should requirements be written at different levels of detail? high

ليه لازم نكتب المتطلبات بمستويات تفصيل مختلفة؟

A. Because natural language cannot express functional requirements
B. Because different types of readers use them in different ways
C. Because every contract legally requires two separate documents
D. Because non-functional requirements cannot appear in the same document as functional ones

ة الصحيحة: B. Because different types of readers use them in different ways

يمة: ين، مهندسين، مطورين) وكل نوع يستخدمها بطريقة. باقي الاختيارات ادعاءات مش مذكورة في المحاضرة.

جع: Chapter 4 — Introduction (Figure 4.2)

6. Which of the following is a definition of system stakeholders? high

ـ الـ Stakeholders؟

A. Only the end-users who interact directly with the system
B. Anyone who is affected by the system in some way and so has a legitimate interest in it
C. Only the customers who pay for the development of the system
D. Only the developers and managers of the contractor

ة الصحيحة: B. Anyone who is affected by the system in some way and so has a legitimate interest in it

يه صح: الـ Stakeholders هم أي حد يتأثر بالنظام وله اهتمام مشروع به (مستخدمين، مديرين، جهات تنظيمية...). الاختيارات الباقية بتضيّق التعر

جع: Chapter 4 — Introduction

7. Which statement describes FUNCTIONAL requirements? high

فية (Functional requirements)؟

A. Constraints on timing, the development process and standards imposed on the system
B. Requirements that apply only to the system as a whole, such as reliability
C. Statements of services the system should provide, how it should react to particular inputs, and how it should behave in particular situations
D. Requirements derived only from laws and regulations

ة الصحيحة: C. Statements of services the system should provide, how it should react to particular inputs, and how it should behave in particular situations

يه صح: الوظيفية = خدمات النظام وردود أفعاله وسلوكه. الاختيار الأول والثاني تعريف الـ Non-functional.

جع: Chapter 4 — Section 4.1

8. Non-functional requirements often apply to: high

لمتطلبات غير الوظيفية غالبًا بتنطبق على:

A. Only the individual services offered to the users
B. Only the user interface of the system
C. Only the database and data representation layer
D. The system as a whole rather than to individual system features or services

ة الصحيحة: D. The system as a whole rather than to individual system features or services

يه صح: الـ NFR بتخص النظام ككل (زي الأداء والاعتمادية). الاختيار الأول غلط لأنه وصف للمتطلبات الوظيفية.

جع: Chapter 4 — Section 4.1

9. Why are non-functional requirements often more critical than individual functional requirements? high

ة الفرد؟NFR

A. Failing to meet a non-functional requirement can make the whole system unusable, while users can usually work around a function that does not meet their needs
B. Because they are always cheaper to verify than functional requirements
C. Because they can always be implemented in a single component
D. Because customers find it easier to express them as measurable numbers

ة الصحيحة: A. Failing to meet a non-functional requirement can make the whole system unusable, while users can usually work around a function that does not meet their needs

يه صح: لأن فشل NFR (زي الاعتمادية في طيارة) ممكن يخلي النظام كله غير صالح، بينما وظيفة واحدة ممكن نلف حواليها. الباقي عكس اللي فو

جع: Chapter 4 — Section 4.1.2

10. According to Figure 4.3, non-functional requirements are classified into: high

كل 4.3، المتطلبات غير الوظيفية بتتقسم إلى:

A. Product, process and operational requirements
B. User, system and domain requirements
C. Product, organizational and external requirements
D. Functional, organizational and legislative requirements

ة الصحيحة: C. Product, organizational and external requirements

يه صح: التصنيف الصحيح: Product و Organizational و External. الاختيار الثاني مخلوط بمفاهيم مش من الشكل.

جع: Chapter 4 — Section 4.1.2 (Figure 4.3)

11. The Mentcare requirement that the system must be available Mon–Fri, 08:30–17:30 with downtime not exceeding 5 seconds a day is an example of a: high

احة Mentcare (الإتاحة من 8:30 لـ 17:30وDowntime لا يزيد عن 5 ثواني) هو مثال على:

A. Organizational requirement
B. External requirement
C. Functional requirement
D. Product requirement

ة الصحيحة: D. Product requirement

يه صح: الإتاحة بتحدد سلوك النظام وقت التشغيل فهي Product requirement. ما بتقولش وظيفة معينة فمش Functional.

جع: Chapter 4 — Section 4.1.2 (Figure 4.4)

12. Which type of non-functional requirement is derived from policies and procedures in the customer's and developer's organizations? high

والمطوّر؟

A. Organizational requirements
B. Product requirements
C. External requirements
D. Domain requirements

ة الصحيحة: A. Organizational requirements

يه صح: الـ Organizational requirements (زي معايير التطوير ولغة البرمجة وطريقة الدخول بالكارت). الـ External بتيجي من عوامل خارج

جع: Chapter 4 — Section 4.1.2

13. Regulatory, legislative and ethical requirements belong to which category of non-functional requirements? high

لأخلاقية تنتمي إلى أي نوع؟

A. Product requirements
B. Organizational requirements
C. Dependability requirements only
D. External requirements

ة الصحيحة: D. External requirements

يه صح: هذه كلها مصدرها خارج النظام وعملية تطويره، فهي External requirements.

جع: Chapter 4 — Section 4.1.2 (Figure 4.3)

14. Whenever possible, non-functional requirements should be written: high

كن، المتطلبات غير الوظيفية يجب أن تُكتب:

A. Quantitatively, so that they can be objectively tested
B. As general goals such as ease of use
C. Only in natural language without any numbers
D. As part of the user interface design

ة الصحيحة: A. Quantitatively, so that they can be objectively tested

يه صح: تُكتب بشكل كمّي قابل للقياس والاختبار. الأهداف العامة (Goals) تترك مجال للتفسير والخلاف.

جع: Chapter 4 — Section 4.1.2

15. Which metric is used to specify the RELIABILITY of a system? high

دية (Reliability)؟

A. Training time
B. Mean time to failure
C. Processed transactions per second
D. Percentage of target dependent statements

ة الصحيحة: B. Mean time to failure

يه صح: Mean time to failure من مقاييس الاعتمادية. Training time للـ Ease of use، والـ transactions/second للـ Speed، والنسبة الـ

جع: Chapter 4 — Section 4.1.2 (Figure 4.5)

16. Which pair of metrics is used to specify EASE OF USE? high

دام (Ease of use)؟

A. Screen refresh time and user/event response time
B. Megabytes and number of ROM chips
C. Time to restart after failure and probability of data corruption
D. Training time and number of help frames

ة الصحيحة: D. Training time and number of help frames

يه صح: Ease of use = Training time + Number of help frames. الاختيار الأول مقاييس Speed، والثاني Size، والثالث Robustness.

جع: Chapter 4 — Section 4.1.2 (Figure 4.5)

17. What does CONSISTENCY mean for a functional requirements specification? high

متطلبات الوظيفية؟ Consistency

A. All services and information required by the user should be defined
B. Requirements should not be contradictory
C. Every requirement should be objectively testable
D. Every requirement should be written in natural language

ة الصحيحة: B. Requirements should not be contradictory

يه صح: Consistency = عدم تناقض المتطلبات. الاختيار الأول هو تعريف Completeness (الأقرب للتشويش).

جع: Chapter 4 — Section 4.1.1

18. Which of the following is NOT one of the three key questions that a feasibility study should answer? high

وى (Feasibility study)؟

A. Does the system contribute to the overall objectives of the organization?
B. Can the system be implemented within schedule and budget using current technology?
C. Have all stakeholders agreed on every detailed requirement?
D. Can the system be integrated with other systems that are used?

ة الصحيحة: C. Have all stakeholders agreed on every detailed requirement?

يه صح: ظام في أهداف المنظمة، إمكانية التنفيذ في الميزانية والجدول بتقنية حالية، والتكامل مع الأنظمة الأخرى. الاتفاق على كل المتطلبات مش منه

جع: Chapter 4 — Introduction (Feasibility studies box)

19. Which are the three key activities of the requirements engineering process? high

ما الأنشطة الثلاثة الأساسية في عملية هندسة المتطلبات؟

A. Elicitation and analysis, specification, and validation
B. Elicitation, design, and implementation
C. Specification, testing, and maintenance
D. Elicitation, validation, and deployment

ة الصحيحة: A. Elicitation and analysis, specification, and validation

يه صح: الأنشطة: Elicitation & analysis، Specification، Validation. باقي الاختيارات فيها مراحل من دورة تطوير البرمجيات مش من RE.

جع: Chapter 4 — Section 4.2

20. How is the requirements engineering process presented in Figure 4.6? high

كل 4.6؟

A. As a strictly sequential process where each activity finishes before the next starts
B. As a one-pass process that ends with system implementation
C. As a process where validation is done only once at the very end
D. As an iterative spiral in which elicitation, specification and validation are interleaved

ة الصحيحة: D. As an iterative spiral in which elicitation, specification and validation are interleaved

يه صح: الشكل Spiral: الأنشطة متداخلة وتتكرر. الاختيارات الأخرى بتصور العملية خطية، والمحاضرة بتؤكد إنها iterative.

جع: Chapter 4 — Section 4.2 (Figure 4.6)

21. Which of the following is NOT a reason why eliciting requirements from stakeholders is difficult? high

ات (Elicitation)؟

A. Stakeholders often do not know what they want except in general terms
B. Political factors may influence the requirements of the system
C. The economic and business environment changes during the analysis
D. Stakeholders always know exactly what they want and express it in precise technical terms

ة الصحيحة: D. Stakeholders always know exactly what they want and express it in precise technical terms

يعوضحش: سببها إن المعنيين غالبًا مش عارفين بالظبط عايزين إيه. الاختيار الصحيح عكس ده تمامًا، فهو ليس سببًا.

جع: Chapter 4 — Section 4.3

22. What is the correct order of the activities in the requirements elicitation and analysis process (Figure 4.7)? high

ـ الـ Elicitation and Analysis (الشكل 4.7)؟

A. Classification and organization → discovery and understanding → documentation → prioritization and negotiation
B. Discovery and understanding → prioritization and negotiation → classification and organization → documentation
C. Discovery and understanding → classification and organization → prioritization and negotiation → documentation
D. Documentation → discovery and understanding → classification and organization → prioritization and negotiation

ة الصحيحة: C. Discovery and understanding → classification and organization → prioritization and negotiation → documentation

يه صح: الترتيب: Discovery → Classification → Prioritization & negotiation → Documentation. الاختيار الثاني بيقدّم الـ oritization لط.

جع: Chapter 4 — Section 4.3 (Figure 4.7)

23. Which activity of the elicitation and analysis process is concerned with finding and resolving conflicts between stakeholders' requirements? high

أي نشاط مسؤول عن اكتشاف وحل التعارضات بين متطلبات أصحاب المصلحة؟

A. Requirements classification and organization
B. Requirements prioritization and negotiation
C. Requirements discovery and understanding
D. Requirements documentation

ة الصحيحة: B. Requirements prioritization and negotiation

يه صح: الـ Prioritization and negotiation هو اللي بيحل التعارض عن طريق التفاوض. الـ Classification بيجمّع المتطلبات المرتبطة فقط.

جع: Chapter 4 — Section 4.3

24. What is a VIEWPOINT in requirements analysis? high

جليل المطلوب؟ Viewpoint

A. A working prototype shown to the stakeholders for evaluation

B. A way of collecting and organizing a set of requirements from a group of stakeholders who have something in common

C. A diagram of the system architecture showing the subsystems

D. A meeting where stakeholders negotiate requirement conflicts

ة الصحيحة: B. A way of collecting and organizing a set of requirements from a group of stakeholders who have something in common

يه صح: الـ Viewpoint = تجميع متطلبات مجموعة stakeholders بينهم قاسم مشترك. الاجتماع التفاوضي نشاط مختلف.

جع: Chapter 4 — Section 4.3 (Viewpoints box)

25. What are the two fundamental approaches to requirements elicitation? high

ما الأسلوبان الأساسيان لاستنباط المتطلبات؟

A. Prototyping and testing

B. Reviews and feasibility studies

C. Interviewing and observation (ethnography)

D. Scenarios and user stories

ة الصحيحة: C. Interviewing and observation (ethnography)

يه صح: الأساسيان: المقابلات (Interviewing) والملاحظة (Ethnography). الـ Stories/Scenarios أدوات لالتقاط المعلومات وليست الأسلوبين

جع: Chapter 4 — Section 4.3.1

26. What is a CLOSED interview? high

لقة (Closed interview)؟

A. An interview with no predefined agenda where a range of issues is explored

B. An interview where the analyst watches people doing their job

C. An interview held only with external stakeholders

D. An interview where the stakeholder answers a predefined set of questions

ة الصحيحة: D. An interview where the stakeholder answers a predefined set of questions

يه صح: المغلقة = أسئلة محددة مسبقًا. الاختيار الأول وصف للمقابلة المفتوحة (Open).

جع: Chapter 4 — Section 4.3.1.1

27. Interviews are NOT an effective technique for eliciting knowledge about: high

المقابلات ليست فعّالة في استنباط المعرفة عن:

A. Organizational requirements and constraints

B. The overall understanding of what stakeholders do

C. How stakeholders might interact with the new system

D. The difficulties stakeholders face with current systems

ة الصحيحة: A. Organizational requirements and constraints

يه صح: علاقات قوة خفية والناس مترددة تتكلم عن السياسة والتنظيم الحقيقي. الباقي هي نقاط قوة المقابلات.

جع: Chapter 4 — Section 4.3.1.1

28. Which of the following does the chapter recommend to get an interview discussion going? high

ما الذي توصي به المحاضرة لبدء النقاش في المقابلة؟

A. A springboard question, a requirements proposal, or working together on a prototype

B. Asking the stakeholder to simply "tell me what you want"

C. Handing the stakeholder the complete formal specification

D. Running a completely open-ended discussion with no questions

ة الصحيحة: A. A springboard question, a requirements proposal, or working together on a prototype

يه صح: نستخدم springboard question أو requirements proposal أو prototype. قول "قولّي عايز إيه" مش بيدّي معلومات مفيدة.

جع: Chapter 4 — Section 4.3.1.1

29. What is ethnography in requirements elicitation? high

[illegible]؟Ethnography

A. A structured interview with a predefined set of questions

B. An observational technique where the analyst immerses himself or herself in the working environment where the system will be used

C. A technique for grouping requirements into coherent clusters

D. A technique for writing high-level narrative descriptions of system use

ة الصحيحة: B. An observational technique where the analyst immerses himself or herself in the working environment where the system will be used

يه صح: الـ Ethnography ملاحظة المحلل لبيئة العمل الفعلية. الاختيار الأول مقابلة مغلقة، والأخير عن Stories.

جع: Chapter 4 — Section 4.3.1.2

30. Ethnography is particularly effective for discovering which two types of requirements? high

[illegible]؟Ethnography

A. Requirements derived from formal business process definitions and from legislation

B. Organizational requirements and domain requirements

C. Requirements for innovative new products and for regulatory approval

D. Requirements derived from how people actually work, and from cooperation and awareness of other people's activities

ة الصحيحة: D. Requirements derived from how people actually work, and from cooperation and awareness of other people's activities

يه صح: تكتشف طريقة العمل الفعلية (مش الرسمية) والتعاون والوعي بأنشطة الآخرين. الاختيارات الأخرى هي أشياء الـ Ethnography ضعيفة ف

جع: Chapter 4 — Section 4.3.1.2

31. What is the main difference between STORIES and SCENARIOS? high

ا الـ Stories والـ Scenarios؟

A. Stories are structured with inputs and outputs, whereas scenarios are narrative high-level text

B. Stories describe only errors, whereas scenarios describe only the normal flow

C. Stories are narrative high-level descriptions of system use, whereas scenarios are structured descriptions with specific information such as inputs and outputs

D. Stories are written by developers, whereas scenarios are written by users

ة الصحيحة: C. Stories are narrative high-level descriptions of system use, whereas scenarios are structured descriptions with specific information such as inputs and outputs

يه صح: الـ Stories وصف عام، والـ Scenarios مهيكلة وأكثر تفصيلًا. الاختيار الأول معكوس.

جع: Chapter 4 — Section 4.3.2

32. Which of the following is NOT a part of a general scenario? high

ليس جزءًا من الـ Scenario؟

A. An estimate of the cost of implementing the scenario
B. A description of the normal flow of events
C. A description of what can go wrong and how problems are handled
D. A description of the system state when the scenario ends

الإجابة الصحيحة: A. An estimate of the cost of implementing the scenario

التوضيح: عناصر الـ Scenario: الافتراض الأولي، التدفق الطبيعي، ما قد يسوء، الأنشطة المتزامنة، حالة النظام عند النهاية. تقدير التكلفة ليس منها.

المرجع: Chapter 4 — Section 4.3.2

33. How should USER requirements be written? high

كيف تُكتب متطلبات المستخدم؟

A. In natural language with simple tables, forms and intuitive diagrams, without software jargon or formal notations
B. Using formal mathematical notations to avoid ambiguity
C. Using software jargon and structured notations that developers understand
D. Using system architecture and design details

الإجابة الصحيحة: A. In natural language with simple tables, forms and intuitive diagrams, without software jargon or formal notations

التوضيح: بلغة طبيعية وجداول ورسومات بسيطة لأن قراءها غير تقنيين. الصيغ الرسمية والجارجون خاصة بمتطلبات النظام.

المرجع: Chapter 4 — Section 4.4

34. If an organization decides that an off-the-shelf software product can meet its needs, the focus should be on developing: medium

لمنظمة أن منتجًا جاهزًا يلبي احتياجاتها، يجب التركيز على:

A. A detailed functional specification
B. Domain requirements
C. Product requirements
D. Information requirements

الإجابة الصحيحة: D. Information requirements

التوضيح: في هذه الحالة التفصيل الوظيفي بلا فائدة، والأفضل تحديد Information requirements (المعلومات اللازمة وطريقة تقديمها).

المرجع: Chapter 4 — Section 4.1.1

35. Domain requirements are derived from: medium

الـ Domain تُشتق من:

A. The policies of the developing organization only
B. The interviews conducted with end-users only
C. The application domain of the system rather than from the specific needs of system users
D. The budget constraints of the customer

الإجابة الصحيحة: C. The application domain of the system rather than from the specific needs of system users

التوضيح: مصدرها مجال تطبيق النظام نفسه وليس احتياجات المستخدمين المحددة. المشكلة أن المهندسين قد لا يفهمون الـ domain.

المرجع: Chapter 4 — Section 4.1.1 (Domain requirements box)

36. How does combining ethnography with prototyping help (Figure 4.8)? medium

كيف يساعد الجمع بين الـ Ethnography والـ Prototyping (الشكل 4.8)؟

A. It removes the need for any stakeholder meetings

B. It guarantees that all organizational requirements are discovered

C. Ethnography informs the prototype so fewer refinement cycles are needed, and prototyping focuses the ethnography on specific questions

D. It replaces the need for requirements validation

ة الصحيحة: C. Ethnography informs the prototype so fewer refinement cycles are needed, and prototyping focuses the ethnography on specific questions

يه صح: الـ Ethnography تقلل دورات تحسين الـ prototype، والـ prototype يوجّه الـ ethnography لأسئلة محددة. باقي الاختيارات مبالغات

جع: Chapter 4 — Section 4.3.1.2 (Figure 4.8)

37. What does the Nokia versus Apple example show about ethnography? medium

ثال Nokia مقابل Apple عن الـ Ethnography؟

A. Ethnography helps in understanding existing systems but does not always help with innovation

B. Ethnography is the best technique for discovering innovative product ideas

C. Ethnography should never be used for new product development

D. Ethnography is effective for discovering broad organizational requirements

ة الصحيحة: A. Ethnography helps in understanding existing systems but does not always help with innovation

يه صح: تساعد في فهم النظم الحالية لكن لا تضمن الابتكار (Apple تجاهلت الاستخدام الحالي وابتكرت). الاختيار الأول عكس المثال.

جع: Chapter 4 — Section 4.3.1.2

Part 2 — Essay

1. Define requirements engineering, and distinguish between user requirements and system requirements. Use the Mentcare monthly report example. high

.Mentcare ثال

Model answer:

Requirements engineering (RE) is the process of finding out, analyzing, documenting and checking the services that a system should provide and the constraints on its operation.

User requirements: statements, in natural language plus diagrams, of the services the system is expected to provide to users and the constraints under which it must operate. They are high-level and general.

System requirements: more detailed descriptions of the system's functions, services and operational constraints. They define exactly what is to be implemented (sometimes called a functional specification) and may be part of the contract.

Example: the user requirement "The Mentcare system shall generate monthly management reports showing the cost of drugs prescribed by each clinic" is expanded into several system requirements (1.1 to 1.5), e.g. generate the report on the last working day after 17:30, one report per clinic, separate reports per dose unit, and restricted access.

They are written differently because different readers use them differently (managers and end-users vs. architects and developers).

جع: Chapter 4 — Introduction (Figures 4.1, 4.2)

2. Compare functional and non-functional requirements. Give an example of each from the Mentcare system. high

.Mentcare لام

Model answer:

Functional requirements: statements of services the system should provide, how it should react to particular inputs, and how it should behave in particular situations (sometimes also what it should not do). Example: "A user shall be able to search the appointments lists for all clinics."

Non-functional requirements: constraints on the services or functions offered by the system, such as timing constraints, constraints on the development process, and constraints imposed by standards. They often apply to the system as a whole. Example: the system shall be available Mon-Fri 08:30-17:30 with downtime not exceeding 5 seconds a day.

Non-functional requirements are often more critical: failing to meet one can make the whole system unusable.

The distinction is not clear-cut: a security requirement (limit access to authorized users) can generate functional requirements such as user authentication facilities.

جع: Chapter 4 — Section 4.1

3. Draw and label the classification of non-functional requirements (Figure 4.3), and give an example of each type. high

كل 4.3) مع مثال لكل نوع.

Model answer:

Non-functional requirements are divided into three branches:

1) Product requirements (performance, space, usability, efficiency, dependability, security). They specify or constrain runtime behavior. Example: availability and allowed downtime.

2) Organizational requirements (operational, development, environmental). They come from policies and procedures of the customer's and developer's organizations. Example: users identify themselves using their health authority identity card.

3) External requirements (regulatory, ethical, legislative; safety/security and accounting sub-types). They come from factors external to the system and its development process. Example: implementing the patient privacy provisions of a national privacy standard.

جع: Chapter 4 — Section 4.1.2 (Figures 4.3, 4.4)

4. Explain why non-functional requirements should be written quantitatively. Rewrite the goal "The system should be easy to use by medical staff" as a testable requirement and list metrics for non-functional properties. high

طلبات غير الوظيفية بشكل كمّي. أعد كتابة الهدف "النظام سهل الاستخدام" كمتطلب قابل للاختبار واذكر مقاييس الخصائص غير الوظيفية.

Model answer:

Stakeholders often propose goals (ease of use, fast response, recovery from failure). Goals leave scope for interpretation and dispute after delivery, and cannot be objectively verified.

Quantitative requirements can be objectively tested during system testing (e.g. by instrumenting the software to count errors).

Testable version: "Medical staff shall be able to use all the system functions after two hours of training. After this training, the average number of errors made by experienced users shall not exceed two per hour of system use."

Metrics (Figure 4.5): Speed = transactions/second, response time, screen refresh time; Size = megabytes, number of ROM chips; Ease of use = training time, number of help frames; Reliability = mean time to failure, probability of unavailability, rate of failure occurrence, availability; Robustness = time to restart after failure, percentage of events causing failure, probability of data corruption on failure; Portability = percentage of target dependent statements, number of target systems.

Difficulties: customers struggle to translate goals into measurable requirements, some goals (e.g. maintainability) have no simple metrics, and verification can be costly.

جع: Chapter 4 — Section 4.1.2 (Figure 4.5)

5. Explain the stages of the requirements elicitation and analysis process (Figure 4.7). high

كل 4.7).

Model answer:

It is an iterative process with continual feedback between activities; the cycle starts with discovery and ends with documentation, and understanding improves with each round.

1) Requirements discovery and understanding: interacting with stakeholders to discover their requirements, including domain requirements and documentation.

2) Requirements classification and organization: grouping related requirements into coherent clusters.

3) Requirements prioritization and negotiation: prioritizing requirements and resolving conflicts between stakeholders through negotiation and compromise.

4) Requirements documentation: documenting the requirements (draft document, whiteboards, wikis or shared documents) as input to the next round of the spiral, using simple language and diagrams.

جع: Chapter 4 — Section 4.3 (Figure 4.7)

6. List and explain the reasons why eliciting and understanding requirements from stakeholders is difficult. high

واشرح أسباب صعوبة استنباط وفهم المتطلبات من أصحاب المصلحة.

Model answer:

1) Stakeholders often do not know what they want except in general terms, find it hard to articulate it, and may make unrealistic demands.

2) Stakeholders express requirements in their own terms with implicit knowledge of their work, which engineers without domain experience may not understand.

3) Different stakeholders with diverse requirements express them differently; engineers must find all sources and discover commonalities and conflicts.

4) Political factors may influence requirements, e.g. managers demanding requirements that increase their influence.

5) The economic and business environment is dynamic; requirements' importance changes and new stakeholders may appear.

جع: Chapter 4 — Section 4.3

7. Compare interviewing and ethnography as requirements elicitation techniques, mentioning the advantages and disadvantages of each. high

والـ Ethnography كأساليب لاستنباط المتطلبات مع ذكر المميزات والعيوب.

Model answer:

Interviewing: talking to people about what they do. Types: closed (predefined questions) and open (no predefined agenda); in practice a mixture.

Advantages of interviews: good for an overall understanding of what stakeholders do, how they might use the new system and their difficulties with current systems; people like talking about their work.

Disadvantages of interviews: stakeholders rarely give specific detailed requirements without a prototype; domain jargon and tacit knowledge are hard to elicit; not effective for organizational requirements because of power relationships and reluctance to discuss politics; may miss essential information.

Ethnography: an analyst immerses himself or herself in the working environment and observes day-to-day work.

Advantages of ethnography: discovers implicit requirements reflecting how people actually work (not the formal process) and requirements from cooperation and awareness of others' activities; reveals critical process details missed by other techniques.

Disadvantages of ethnography: focused on the end-user, so not effective for broader organizational or domain requirements or for innovation.

Conclusion: use a mix of both, with other techniques.

جع: Chapter 4 — Section 4.3.1

8. What is the difference between stories and scenarios? List what a scenario may include. high

ما الفرق بين القصص والسيناريوهات؟ اذكر ما قد يتضمنه الـ Scenario.

Model answer:

Stories and scenarios are essentially the same: a description of how the system can be used for a particular task. The difference is in structure and level of detail.

Stories: narrative text, high-level description of system use; good for setting out the "big picture" and reaching a wider community (e.g. the iLearn photo-sharing story).

Scenarios: structured descriptions of example user interaction sessions with specific information such as inputs and outputs; more detailed and used to propose requirements.

A scenario may include: (1) what the system and users expect when it starts, (2) the normal flow of events, (3) what can go wrong and how it is handled, (4) other activities going on at the same time, (5) the system state when the scenario ends.

المرجع: Chapter 4 — Section 4.3.2 (Figures 4.9, 4.10)

9. What is a feasibility study? State the three key questions it should answer. high

ما هي دراسة الجدوى؟ اذكر الأسئلة الثلاثة الرئيسية التي يجب أن تجيب عنها.

Model answer:

A feasibility study is a short, focused study early in the RE process that assesses whether the system is technically and financially feasible, helping management decide whether to go ahead with procurement or development.

Key questions: (1) Does the system contribute to the overall objectives of the organization? (2) Can the system be implemented within schedule and budget using current technology? (3) Can the system be integrated with other systems that are used?

If the answer to any of these is no, you should probably not go ahead with the project.

المرجع: Chapter 4 — Introduction (Feasibility studies box)

10. Explain the spiral view of the requirements engineering process (Figure 4.6). medium

اشرح العرض الحلزوني (Spiral) لعملية هندسة المتطلبات (الشكل 4.6).

Model answer:

RE involves three key activities: elicitation and analysis, specification, and validation. In practice they are interleaved and organized as an iterative spiral; the output is a system requirements document.

Effort on each activity depends on the stage of the process, the type of system and the available budget.

Early iterations focus on high-level business requirements and user requirements; outer rings focus on more detailed system requirements and non-functional requirements.

The number of iterations can vary, so the spiral can be exited after some or all user requirements are elicited. Agile development can be used instead of prototyping so requirements and implementation are developed together.

Requirements change in virtually all systems, so changes must be managed (requirements management).

المرجع: Chapter 4 — Section 4.2 (Figure 4.6)

11. Who are the system stakeholders? Give examples of Mentcare stakeholders. medium

من هم أصحاب المصلحة في النظام؟ أعط أمثلة على أصحاب المصلحة في Mentcare.

Model answer:

System stakeholders include anyone affected by the system in some way and so anyone with a legitimate interest in it: end-users, managers, and external stakeholders such as regulators who certify acceptability.

Mentcare examples: patients and their relatives; doctors; nurses; medical receptionists; IT staff; a medical ethics manager; health care managers; medical records staff.

المرجع: Chapter 4 — Introduction

12. Explain why it is only possible to achieve completeness and consistency for very small systems. medium

, الـ Completeness والـ Consistency فقط للأنظمة الصغيرة جدًا.

Model answer:

Completeness means all services and information required by the user are defined; consistency means requirements are not contradictory.

Reason 1: it is easy to make mistakes and omissions when writing specifications for large, complex systems.

Reason 2: large systems have many stakeholders with different backgrounds and expectations, whose needs are often inconsistent. These inconsistencies may only be discovered after deeper analysis or during development.

Imprecision can also lead to disputes: developers may interpret an ambiguous requirement in the way that is easiest to implement (e.g. the Mentcare search requirement).

جع: Chapter 4 — Section 4.1.1

13. Explain why non-functional requirements are difficult to implement and trace, and how they relate to functional requirements. medium

لماذا يصعب تنفيذ وتتبع المتطلبات غير الوظيفية وكيف ترتبط بالمتطلبات الوظيفية.

Model answer:

Their implementation may be spread throughout the system, for two reasons:

1) They may affect the overall architecture rather than individual components (e.g. organizing the system to minimize communication between components to meet performance requirements).

2) A single non-functional requirement, such as security, may generate several related functional requirements (new services) and may also constrain existing requirements, e.g. by limiting access to information.

Non-functional requirements also conflict and interact with others (e.g. a card-reader identification requirement conflicts with mobile access from tablets or smartphones).

Ideally, highlight requirements related to emergent properties such as performance or reliability, e.g. in a separate section.

جع: Chapter 4 — Section 4.1.2

14. Describe how ethnography can be combined with prototyping, and discuss its limitation for innovation (Nokia vs Apple). medium

الـ Ethnography مع الـ Prototyping، وناقش حدودها في الابتكار (Nokia مقابل Apple).

Model answer:

Figure 4.8: ethnographic analysis informs generic system development and prototyping, so fewer prototype refinement cycles are needed. Prototype evaluation identifies problems and questions that are discussed in debriefing meetings and then guide focused ethnography in the next phase.

Limitation: ethnography helps to understand existing systems but not always to innovate. Nokia is said to have used ethnography to see how people used phones, whereas Apple ignored current use and revolutionized the industry with the iPhone.

Because of its end-user focus it is not effective for broader organizational or domain requirements, so use it with other techniques.

جع: Chapter 4 — Section 4.3.1.2 (Figure 4.8)

15. What makes an effective interviewer? Mention the difficulties of eliciting domain knowledge in interviews. medium

جال (Domain knowledge) في المقابلات.

Model answer:

An effective interviewer is open-minded, avoids preconceived ideas, and is willing to change his or her mind if surprising requirements appear; and prompts the interviewee with a springboard question, a requirements proposal, or a prototype instead of "tell me what you want".

Difficulties with domain knowledge: (1) specialists use jargon in a precise and subtle way that engineers may misunderstand; (2) some knowledge is so familiar that stakeholders find it hard to explain or think it is not worth mentioning (e.g. the librarian cataloguing every acquisition).

Interviewing alone may miss essential information, so combine it with other techniques.

جع: Chapter 4 — Section 4.3.1.1